NOTE

I use the terms *claims* and *attributes* interchangeably when describing the data in verifiable credentials. The W3C specification uses the word *claim* to emphasize that the issuer is *asserting* something about the holder. In the context of authorization, these claims serve as attributes that the policy engine can use to inform its authorization decision. You can think of claims as the attributes contained in the credential, and we will use whichever term is most natural in a given context.

16.2.1 Credential presentation

The trust triangle in figure 16.1 shows how verifiable credentials work in practice. A trusted issuer makes claims about a holder and signs them so they can be verified by a relying party. When a holder presents a credential, the verifier can use cryptography to confirm four things: who the issuer is, that the credential has not been tampered with, that it was issued to the wallet presenting it, and that it has not been revoked. The holder transports and presents the credential, but is not the source of trust. When ACME verifies a credential during an access request, the key question is not whether the holder is honest. Instead, the question is whether the issuer signed the credential and whether the claims meet the policy's requirements.

NOTE

The previous paragraph was careful to specify that cryptography guarantees the credential was issued to a specific wallet, not a person. The link between the wallet and the person presenting it is established outside the credential itself. Most wallets rely on the device's biometric unlock, such as a fingerprint or face scan, to ensure that only the device's rightful controller can present a credential. Issuers also help reinforce this link by performing identity proofing when issuing the credential, confirming that the claims in the credential belong to the person who controls the wallet it is linked to.